

Ramanuja Magadi Anantha Krishna

Data Scientist | Data Engineer | Agentic Engineer

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SUMMARY

Data Engineer and Scientist with 2+ years of professional experience and a Master of Science in Data Science, Analytics, and Engineering from Arizona State University (**GPA 4.0/4.0**). Resolved 950+ enterprise production tickets at Informatica while sustaining customer satisfaction above 4.65/5.0 and cutting recurring incident resolution time by approximately 30%. Skilled in Agentic AI, Python, SQL, statistical analysis, predictive modeling, machine learning, NLP, and Retrieval Augmented Generation (RAG), with hands-on experience deploying ETL pipelines, real-time streaming, and cloud architectures on AWS, GCP, and Azure.

WORK EXPERIENCE

Technical Events Aide, *Project Humanities, Arizona State University, Tempe, Arizona*

Oct 2024 - May 2026

- Performed end-to-end analytics on Hacks for Humanity hackathon records using Stova Analytics and Python, accelerating executive-facing report turnaround by approximately 50% across demographics, engagement KPIs, and post-event outcomes.
- Engineered an algorithmic team formation pipeline for 100+ participants and automated expense reporting workflows, reducing manual processing time by approximately 40% and improving reporting accuracy.

Associate Software Engineer, *Informatica, Bangalore, India*

Oct 2022 - Jun 2024

- Resolved 950+ complex production tickets in Cloud Data Integration (CDI) and Application Integration for Fortune 500 enterprise clients across United States and Europe, sustaining customer satisfaction above 4.65.
- Diagnosed Java stack traces, network captures, and Informatica agent logs to remediate production outages, stale transaction states, and PowerCenter to IDMC migration failures.
- Automated diagnostic workflows and SQL-based validation checks using Shell scripting and Linux tooling, reducing mean time to resolution by approximately 30% for recurring pipeline incidents.

Software Engineering Intern, *Informatica, Bangalore, India*

Feb 2022 - Oct 2022

- Delivered Cloud Data Integration projects covering connector configuration, schema mapping, and ETL pipeline troubleshooting; authored many technical articles and videos adopted into onboarding documentation.

EDUCATION AND CERTIFICATIONS

MS in Data Science, Analytics, and Engineering, Arizona State University, Tempe, Arizona

Aug 2024 - May 2026

GPA 4.0. Coursework: Data Processing at Scale, Statistical Machine Learning, Data Mining, Advanced Network Security, High Performance Computing, DSA, Computer Organization, OS, DBMS, Web Technology, AI, ML, Big Data Analytics

Bachelor of Engineering in Information Science and Engineering, Visvesvaraya Technological University, India Aug 2018 - Jul 2022 **GPA 3.91**

Claude 101, Google Cloud Core Services, GCP Essential Training for Developers, Learning Vertex AI: MLOps with Google Cloud, Google Cloud Foundations

PROJECTS, RESEARCH & OPEN SOURCE CONTRIBUTIONS

Agentic AI Systems and Comprehensive Knowledge Base — From No Agent LLM to Lang Graph [GitHub]

May 2026

- Built ReAct agents in Python with native JSON-Schema tool calling (calculator, lookup, `wikipedia.search/get_article`, sandboxed `python.exec`); token-threshold compression & slicing to preserve `tool.call_id` linking, episodic memory via `text-embedding-3-small` (1536-d, cosine, 0.35 threshold) with lesson-learned, orchestrated pipeline (researcher→writer) and manager-worker - multi-agent (destination/activities researchers + budget analyzer as native tools) and ported the core loop to LangGraph `StateGraph` with reducer state, *Stack: Python, OpenAI/OpenRouter SDK, NumPy, LangGraph, CrewAI, Streamlit*.

Earnings Call Risk Analysis Engine [GitHub]

Jan 2026 - Apr 2026

- Built an Agentic financial intelligence system fusing NLP signals from 1,700+ earnings call transcripts with live market feeds to generate real-time BUY, HOLD, and SELL signals; applied reinforcement learning hyperparameter optimization and SHAP interpretability under strict chronological validation. *Stack: Python, XGBoost, FinBERT, LangGraph, SHAP, yfinance API, Streamlit*.

Interactive Workday Application [GitHub]

Apr 2025

- Developed a conversational LLM agent over the Workday Worker Node API to retrieve employee records and draft context-aware emails, cutting record retrieval time from minutes to seconds; deployed on Ubuntu 22.04 with FAISS semantic retrieval and RocksDB persistence. *Stack: Python, Workday API, FAISS, RocksDB, Qwen2.5-0.5B-Instruct*.

Code RAG: Repository Aware Code Assistant [GitHub]

Dec 2024

- Implemented a CLI Retrieval Augmented Generation system answering code-specific queries from indexed repository context; FAISS dense retrieval, RocksDB persistence, packaged in Docker on Ubuntu 22.04. *Stack: Python, Docker, RocksDB, FAISS, Qwen2.5-Coder*.

Additional: [Air Gesture Keyboard](#) (96% recognition accuracy, published a paper in IJRASET); [Graph Exploration of NYC Taxi Trips](#) - Ingested NYC taxi data via Apache Kafka streaming into Neo4j; containerized inside Docker, orchestrated with Kubernetes & Applied Page Rank; Subreddit Processing (4M+ rows analyzed in PostgreSQL).

TECHNICAL SKILLS

Programming and Query: Python (NumPy, Pandas, Scikit-learn, transformers, torch), SQL, R Programming, Scala, Shell Scripting, Julia, Cypher-shell

Machine Learning and AI: Agentic AI [ReAct agent, Lang graph., Crew AI, Multi-agent conversation, Summarization and Compression agent], XGBoost, FinBERT, SHAP, FAISS, Predictive Modeling, Feature Engineering, Classification, Regression, Clustering, Time Series Analysis, NLP, LLMs, Retrieval Augmented Generation (RAG), Reinforcement Learning, Hypothesis Testing, MLOps, Precision and Recall, ROC-AUC

Analytics & Visualization: Exploratory Analysis (EDA), Mining, Dashboards, Tableau, Streamlit, Matplotlib, Seaborn, Stova Analytics, Business Intelligence

Data Engineering: Apache Spark, Apache Kafka, Neo4j, PostgreSQL, MongoDB, Snowflake, RocksDB, Warehousing, Modeling, ETL and ELT Pipelines, Salesforce Developer.

Cloud and DevOps: AWS (S3, EC2, Redshift), GCP (BigQuery, Vertex AI, Compute Engine), Azure DevOps, Docker, Kubernetes, CI/CD, Linux, Git, GitHub, Postman

Enterprise and Methodologies: Informatica CDI, IDMC, PowerCenter to IDMC Migration, Stakeholder Management, Wireshark.